

ECO387C - Fall 2026

Discussion 4

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1 Overview

Dynamic programming is widely used in theoretical economics, particularly in macro. Consider the canonical neoclassical growth model, where a central planner solves the following maximization problem:

$$\max_{\{c_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to an initial arbitrary k_0 , a non-negativity constraint on capital $k_t \geq 0$, and the resource constraint:

$$c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t.$$

While highly stylized, this problem is extremely useful as a laboratory for testing numerical algorithms because under a certain parameterization (full depreciation, i.e., $\delta = 1$, and log utility, i.e., $u(c_t) = \ln(c_t)$), a **closed-form** solution can be found. When deploying more sophisticated algorithms on richer problems, we can gain some confidence that they are working correctly if their numerical solution to the neoclassical growth model is the same as the analytical solution that we will find later in this discussion.

2 Euler Equation

When the problem is written **sequentially**, we can set up the Lagrangian to find the consumption-Euler equation:

$$u'(c_t) = \beta(f'(k_{t+1}) + 1 - \delta)u'(c_{t+1}).$$

This **characterizes** the solution, i.e., we know that the solution has to satisfy the consumption-Euler equation, but it is not **the** solution. When we recast this problem recursively and use dynamic programming, we can often go further (and answer much richer problems).

We can also find the consumption Euler equation starting directly from dynamic programming. The general strategy is the following recipe:

1. Identify the state and control variables.
2. Set up the Bellman equation.
3. Obtain first-order conditions by taking the derivative w.r.t. the **control** variables.
4. Invoke the Envelope theorem by taking the derivative w.r.t. the **state** variables.

When applied to the neoclassical growth model:

1. The state variable is k_t , and the control variables are c_t and k_{t+1} . To solve the problem, we need to find:
 - The value function $v(k_t)$, which is the value of behaving optimally, taking the state variable k_t as given.
 - The policy functions for next period's capital k_{t+1} and consumption c_t .

However, note that we only need to solve for one of the two control variables (we solve for k_{t+1} below), since the other can be pinned down by the resource constraint.

2. The Bellman equation for this problem is:

$$v(k) = \max_x \{u(k^\alpha + (1 - \delta)k - x) + \beta v(x)\}$$

where x is a dummy variable indexing possible choices for $k' = g(k)$.

While we don't know what form the value function takes explicitly, we do know that it satisfies the Bellman equation above. We will use this to solve for the value function $v(k)$.¹

¹To reduce the notational burden, we'll assume full depreciation, or $\delta = 1$, from now on.

Note that we are able to drop the t subscript because this is a **stationary problem**. In other words, the problem that the agent solves stays the same in each period, so there's nothing different between $t = 0$ and $t = 1$.²

3. Treating $v(x)$ as known, the RHS of the Bellman equation implies the following FOC:

$$u'(f(k) - x) = \beta v'(x)$$

However, we don't know $v(x)$, so we also don't immediately know what $v'(x)$ is.

4. Use the Envelope theorem:

Substituting in the policy $x = g(k)$ into the Bellman equation:

$$v(k) = u(f(k) - g(k)) + \beta v(g(k))$$

Taking the total derivative w.r.t. the state variable gives:

$$\begin{aligned} v'(k) &= u'(f(k) - g(k))f'(k) + \underbrace{[-u'(f(k) - g(k)) + \beta v'(g(k))]}_{=0 \text{ by FOC}} g'(k) \\ &= u'(f(k) - g(k))f'(k) \end{aligned}$$

In other words, the Envelope theorem tells us that the total derivative of the value function w.r.t. k is given by the partial derivative of the objective function w.r.t. k **evaluated at the optimum**.

Evaluating this at the optimum $g(k)$ yields:

$$v'(g(k)) = u'(\underbrace{f(g(k)) - g(g(k))}_{c'})f'(g(k))$$

Combining this with the first-order condition gives us the consumption-Euler equation:

$$u'(c) = \beta f'(k')u'(c')$$

²An example of a problem that is NOT stationary is if this problem has a FINITE horizon.

3 Guess and verify

Now we start to go beyond the Euler equation and try to solve the value function and policy function fully. For simple problems, the guess-and-verify strategy is often effective.

1. Guess that the value function takes the following form:

$$v(k) = E + F \ln(k)$$

2. This guess implies that:

$$v(k) = \max_{\{k'\}} \{ \ln(k^\alpha - k') + \beta [E + F \ln(k')] \}$$

3. Taking the derivative and rearranging yields:

$$k' = \frac{\beta F}{1 + \beta F} k^\alpha$$

4. Substituting this in and rearranging further yields:

$$\begin{aligned} v(k) &= \ln \left(k^\alpha - \frac{\beta F}{1 + \beta F} k^\alpha \right) + \beta \left[E + F \ln \left(\frac{\beta F}{1 + \beta F} k^\alpha \right) \right] \\ &= \underbrace{\beta E + \beta F \ln(\beta F) - (1 + \beta F) \ln(1 + \beta F)}_E + \underbrace{\alpha(1 + \beta F) \ln(k)}_F \end{aligned}$$

5. Finally, equating the coefficients and solving for E and F yields:

$$\begin{aligned} E &= \frac{1}{1 - \beta} \left[\ln(1 - \alpha\beta) + \frac{\alpha\beta}{1 - \alpha\beta} \ln(\alpha\beta) \right] \\ F &= \frac{\alpha}{1 - \alpha\beta} \end{aligned}$$

4 Value function iteration

As noted above, the solutions to the dynamic programming problem are two functions: the value function and the policy function. It turns out that if we iterate on the equation

$$V_{j+1}(k) = \max_{k' \geq 0} \{ u(k^\alpha - k' + k(1 - \delta)) + \beta V_j(k') \}$$

starting from an arbitrary starting function $V_0(k)$, we converge to the true solution (a fixed point). This is called value function iteration. To illustrate this method, assume that $\delta = 1$ and $u(c) = \ln(c)$. Then

$$V_{j+1}(k) = \max_{k' \geq 0} \{ \ln(k^\alpha - k') + \beta V_j(k') \}.$$

We now go through the different iterations.

1. Set $V_0(k) \equiv 0$. This is the function which is zero everywhere. We can find $V_1(k)$ by finding

$$V_1(k) = \max_{k' \geq 0} \{ \ln(k^\alpha - k') \}.$$

The optimal solution is $k' = 0$. Plugging this back in gives

$$V_1(k) = \ln(k^\alpha).$$

2. Now solve the problem

$$\begin{aligned} V_2(k) &= \max_{k' \geq 0} \{ \ln(k^\alpha - k') + \beta V_1(k') \} \\ &= \max_{k' \geq 0} \{ \ln(k^\alpha - k') + \beta \ln((k')^\alpha) \} \end{aligned}$$

Now the FOC is

$$\frac{1}{k^\alpha - k'} = \beta \frac{1}{(k')^\alpha} \alpha (k')^{\alpha-1}$$

or

$$k' = \frac{\beta \alpha}{1 + \beta \alpha} k^\alpha$$

Hence,

$$\begin{aligned} V_2(k) &= \ln\left(\frac{1}{1 + \beta \alpha} k^\alpha\right) + \beta \alpha \ln\left(\frac{\beta \alpha}{1 + \beta \alpha} k^\alpha\right) \\ &= \ln\left(\frac{1}{1 + \beta \alpha}\right) + \beta \alpha \ln\left(\frac{\beta \alpha}{1 + \beta \alpha}\right) + (1 + \beta \alpha) \ln(k^\alpha) \end{aligned}$$

3. Next solve

$$\begin{aligned} V_3(k) &= \max_{k' \geq 0} \{ \ln(k^\alpha - k') + \beta V_2(k') \} \\ &= \max_{k' \geq 0} \left\{ \ln(k^\alpha - k') + \beta \left[\ln\left(\frac{1}{1 + \beta \alpha}\right) + \beta \alpha \ln\left(\frac{\beta \alpha}{1 + \beta \alpha}\right) + (1 + \beta \alpha) \ln((k')^\alpha) \right] \right\} \end{aligned}$$

The first order condition is

$$\frac{1}{k^\alpha - k'} = \beta(1 + \beta\alpha) \frac{1}{(k')^\alpha} \alpha (k')^{\alpha-1}$$

or

$$k' = \frac{\beta\alpha(1 + \beta\alpha)}{1 + \beta\alpha(1 + \beta\alpha)} k^\alpha$$

4. ...³

5. In the limit, we will have

$$k' = \frac{\beta\alpha(1 + \alpha\beta + (\alpha\beta)^2 + \dots)}{1 + \beta\alpha(1 + \alpha\beta + (\alpha\beta)^2 + \dots)} k^\alpha = \beta\alpha k^\alpha$$

We can then write

$$\begin{aligned} \ln(k_t) &= \ln(\beta\alpha k_{t-1}^\alpha) = \ln(\beta\alpha) + \alpha \ln(k_{t-1}) \\ &= \ln(\beta\alpha) + \alpha \ln(\beta\alpha) + \alpha^2 \ln(k_{t-2}) \\ &= \ln(\beta\alpha) + \alpha \ln(\beta\alpha) + \alpha^2 [\ln(\beta\alpha) + \alpha \ln(k_{t-3})] \\ &= (1 + \alpha + \alpha^2) \ln(\beta\alpha) + \alpha^3 \ln(k_{t-3}) \\ &= (1 + \alpha + \alpha^2 + \dots + \alpha^{t-1}) \ln(\beta\alpha) + \alpha^t \ln(k_0) \\ &= \frac{1 - \alpha^t}{1 - \alpha} \ln(\beta\alpha) + \alpha^t \ln(k_0) \end{aligned}$$

To find the associated value function, plug in $\ln(k_t)$:

$$\begin{aligned} V(k) &= \sum_{t=0}^{\infty} \beta^t \ln(k_t^\alpha - k_{t+1}) = \sum_{t=0}^{\infty} \beta^t \ln((1 - \beta\alpha) k_t^\alpha) \\ &= \sum_{t=0}^{\infty} \beta^t \left[\ln(1 - \beta\alpha) + \alpha \left\{ \frac{1 - \alpha^t}{1 - \alpha} \ln(\beta\alpha) + \alpha^t \ln(k) \right\} \right] \\ &= \sum_{t=0}^{\infty} \beta^t \left[\ln(1 - \beta\alpha) + \frac{\alpha}{1 - \alpha} \ln(\beta\alpha) - \frac{\alpha}{1 - \alpha} \alpha^t \ln(\beta\alpha) + \alpha \alpha^t \ln(k) \right] \\ &= \frac{\ln(1 - \beta\alpha) + \frac{\alpha}{1 - \alpha} \ln(\beta\alpha)}{1 - \beta} + \left[\alpha \ln(k) - \frac{\alpha}{1 - \alpha} \ln(\beta\alpha) \right] \frac{1}{1 - \alpha\beta} \\ &= \frac{\ln(1 - \beta\alpha)}{1 - \beta} + \frac{1}{1 - \beta} \frac{\alpha}{1 - \alpha} \ln(\beta\alpha) \left[\frac{\beta(1 - \alpha)}{1 - \alpha\beta} \right] + \frac{\alpha}{1 - \alpha\beta} \ln(k) \\ &= \frac{1}{1 - \beta} \left[\ln(1 - \beta\alpha) + \frac{\beta\alpha}{1 - \alpha\beta} \ln(\beta\alpha) \right] + \frac{\alpha}{1 - \alpha\beta} \ln(k) \end{aligned}$$

³NEVER do this by hand!

5 Other methods

You may notice that in both methods that we covered today, we never actually used the consumption-Euler equation to find the value and policy function. Indeed, there are powerful algorithms that iterate on/use the consumption-Euler equation instead of the Bellman equation, and they can be orders of magnitude faster.

However, iterating on the Bellman equation is still extremely popular for two reasons:

1. Convergence. As long as Blackwell's sufficiency conditions are satisfied, your initial guess of the value function is **guaranteed** to eventually converge to the right solution. Many faster methods do not have this guarantee and can be sensitive to your initial guess.
2. Value function iteration is extremely flexible and robust to the types of problems. Throughout the semester, you will be exposed to problems where the consumption-Euler equation does not always hold, or does not exist.